PROGRAM 3: Engineering Image Generation Using Text Prompts

AIM

To generate engineering-related images from text prompts using a pretrained text-to-image model.

ALGORITHM

1. Import the required libraries.
2. Check GPU availability.
3. Load the pretrained Stable Diffusion Turbo model.
4. Accept a text prompt.
5. Generate an image.
6. Display and save the generated image.

CODE

Important: This program requires a GPU. Select T4 GPU in Colab if available.

```
import torch
from diffusers import AutoPipelineForText2Image
from IPython.display import display

if not torch.cuda.is_available():
    raise RuntimeError("Please enable GPU in Colab settings.")

pipe = AutoPipelineForText2Image.from_pretrained(
    "stabilityai/sd-turbo",
    torch_dtype=torch.float16,
    variant="fp16"
)

pipe = pipe.to("cuda")

prompt = input("Enter image prompt: ")

image = pipe(
```

```
    prompt=prompt,
    num_inference_steps=1,
    guidance_scale=0.0,
    height=512,
    width=512
).images[0]

display(image)

image.save("engineering_image.png")

print("Image generated and saved successfully.")
```

INPUT

Enter image prompt:

A modern robotic arm working in an engineering laboratory

OUTPUT

An AI-generated image of a robotic arm in an engineering laboratory is displayed.

Generated file: engineering_image.png

RESULT

Thus, an engineering-related image was generated successfully using a pretrained text-to-image model.